Test Series Code

13

All India Test Series
NEET (2025-26)

SET

01

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 15/04/2026

SYLLABUS

Physics:	Full Syllabus "As per NTA"
Chemistry:	Full Syllabus "As per NTA"
Botany:	Full Syllabus "As per NTA"
Zoology:	Full Syllabus "As per NTA"

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A plano-convex lens has a central thickness of 2 mm and an aperture diameter of 8 cm. If the speed of light within the lens material is $2.4\times10^8\text{ ms}^{-1}$, what is the approximate focal length of the lens?

(1) 80 cm (2) 160 cm

(3) 120 cm (4) 40 cm

2. Two particles of masses m and $4m$ are moving in horizontal circles. The heavier mass moves in a circle of radius $\frac{R}{4}$, while the lighter mass moves in a circle of radius R. If both particles experience the exact same centripetal force, the speed of the lighter particle is k times the speed of the heavier particle. Find the value of k.

(1) 2 (2) 8

(3) 4 (4) 16

3. A uniform circular hoop of mass $3m$ and radius R rotates freely about its central axis with an initial angular velocity ω_0. Two small beads, each of mass m, are gently attached to diametrically opposite points on its rim. What is the new angular velocity of the system?

(1) $0.6\omega_0$ (2) $0.4\omega_0$

(3) $0.8\omega_0$ (4) $0.5\omega_0$

4. A particle undergoes simple harmonic motion along a straight line with an amplitude of 5 cm. At a displacement of 3 cm from the mean position, the numerical value of its acceleration is exactly equal to the numerical value of its velocity. Determine the time period of the oscillation.

(1) $\frac{3\pi}{2}$ s

(2) $\frac{2\pi}{5}$ s

(3) $\frac{4\pi}{3}$ s

(4) $\frac{5\pi}{4}$ s

5. A nucleus at rest disintegrates into two smaller nuclei with their masses in the ratio of 2:1. After disintegration they will move:

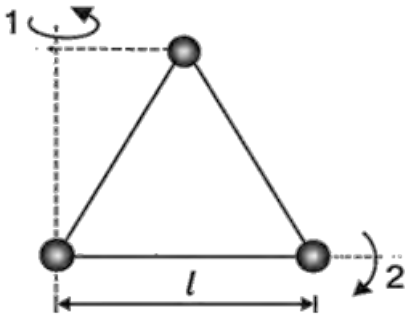
(1) In opposite directions with speed in the ratio of 1 : 2 respectively

(2) In opposite directions with speed in the ratio of 2 : 1 respectively

(3) In the same direction with same speed.

(4) In opposite directions with the same speed.

6. Three particles, each of mass m, are placed at the corners of an equilateral triangle of side l.



Read the following statements and choose the option containing **correct** statements.

(A) Moment of inertia about axis '1' is $\frac{5}{4}ml^2$

(B) Moment of inertia about axis '2' is $\frac{3}{4}ml^2$

(C) Moment of inertia about an axis passing through one corner and perpendicular to the plane is $2ml^2$

(1) A and B only (2) B and C only

(3) A only (4) A, B and C

7. In a YDSE setup, the slit separation is 0.1 mm. A bright fringe is observed at an angular position of 0.018 rad for two distinct wavelengths, λ_1 and λ_2, both falling within the visible spectrum (380 nm to 740 nm). What are the values of these two wavelengths?

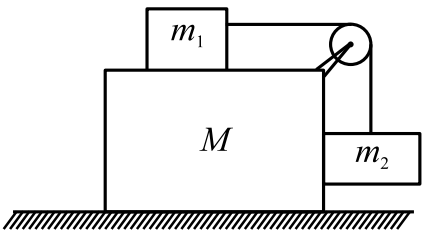
(1) 750 nm, 500 nm

(2) 650 nm, 400 nm

(3) 550 nm, 450 nm

(4) 600 nm, 450 nm

8. In the given arrangement, all surfaces are smooth and string are ideal. What acceleration should be given to the block M so that the block m_2 does not slide down?



(1) $\frac{m_2 g}{m_1}$ (2) $\frac{m_1 g}{m_2}$

(3) g (4) $\frac{m_2 g}{m_1 + m_2}$

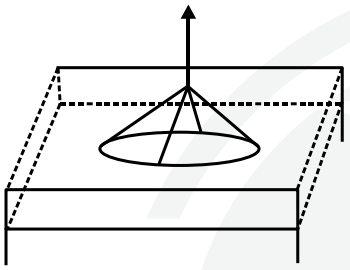
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9. An ideal liquid exhibits steady flow through a horizontal tube of varying cross-section. The widest and narrowest parts of the tube have diameters of $7.5\ \text{cm}$ and $3.0\ \text{cm}$, respectively. Find the ratio of the minimum fluid velocity to the maximum fluid velocity in the tube.

(1) $\frac{25}{4}$ (2) $\frac{4}{25}$

(3) $\frac{2}{5}$ (4) $\frac{5}{2}$

10. The surface tension of water is 75 dyne/cm. The minimum vertical force $\frac{1.17}{x}\ \text{N}$ is required to pull a thin wire ring up if the circumference is $20\ \text{cm}$ and weight is $0.1\ \text{N}$. Find *x*.



(1) 3 (2) 9

(3) 12 (4) 15

11. An elevator has a fully loaded mass of $1200\ \text{kg}$. It moves upward at a constant speed of $2.5\ \text{m s}^{-1}$. If a constant frictional drag of $3000\ \text{N}$ opposes its motion, determine the minimum power the motor must deliver. (Assume $g = 10\ \text{m s}^{-2}$)

(1) 30.0 kW

(2) 42.5 kW

(3) 45.0 kW

(4) 37.5 kW

12. A metallic sphere has a mass of $40\ \text{g}$. If its specific heat capacity is $0.15\ \text{cal g}^{-1}{}^{\circ}\text{C}^{-1}$, find the water equivalent of this sphere.

(1) 10 g (2) 8 g

(3) 12 g (4) 6 g

13. A conducting wire of length l, area A, and resistivity ρ is connected to a battery. If the length of the wire is doubled and the area of cross-section is halved, the resultant current would be:

(1) $\frac{1}{4}\frac{\rho l}{VA}$ (2) $\frac{3}{4}\frac{VA}{\rho l}$

(3) $4\frac{VA}{\rho l}$ (4) $\frac{1}{4}\frac{VA}{\rho l}$

14. An ideal transformer connected to a $200\ \text{V}$ AC supply has 1000 turns in its primary coil and 50 turns in its secondary coil. If a purely resistive load of 2Ω is connected across the secondary coil, match the electrical quantities in **List-I** with their corresponding numerical values in **List-II**.

List-I		**List-II**	
(A)	Secondary Voltage (in V)	(I)	5
(B)	Secondary Current (in A)	(II)	10
(C)	Primary Current (in A)	(III)	50
(D)	Output Power (in W)	(IV)	0.25

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III

(2) A-I, B-IV, C-II, D-III

(3) A-II, B-I, C-IV, D-III

(4) A-IV, B-I, C-III, D-II

15. If $\vec{A}$ and $\vec{B}$ are two vectors satisfying the relation $\vec{A}\cdot\vec{B} = |\vec{A}\times\vec{B}|$, then the value of $|\vec{A}-\vec{B}|$ will be:

(1) $\sqrt{A^2+B^2-\sqrt{2}AB}$

(2) $\sqrt{A^2+B^2}$

(3) $\sqrt{A^2+B^2+\sqrt{2}AB}$

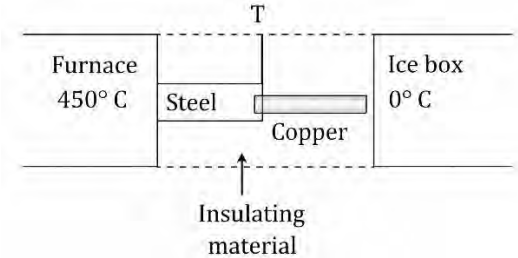
(4) $\sqrt{A^2+B^2+2AB}$

16. Three energy levels L_1, L_2 and L_3 of a hydrogen atom correspond to increasing values of energy i.e., $E_{L1} < E_{L2} < E_{L3}$. If the wavelength corresponding to the transitions $L_3 \to L_2, L_2 \to L_1$ and $L_3 \to L_1$ are λ_3, λ_2 and λ_1 respectively, then:

(1) $\lambda_3 = \lambda_1 + \lambda_2$ (2) $\lambda_1 = \frac{\lambda_2\lambda_3}{\lambda_2+\lambda_3}$

(3) $\lambda_1 = \lambda_2 + \lambda_3$ (4) $\lambda_3 = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

17. Two rods of steel (Rod 1) and copper (Rod 2) are arranged as shown. $\frac{K_2}{K_1} = 9, \frac{A_1}{A_2} = 2, \frac{L_1}{L_2} = 2$. Find the junction temperature T in steady state. (symbols have their usual meaning)



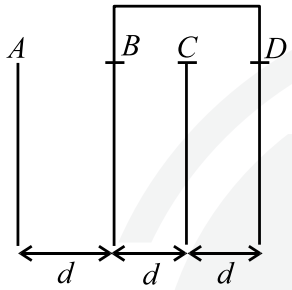
(1) 18°C (2) 14°C

(3) 45°C (4) 150°C

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18. A negative charge is coming towards the observer. The direction of the magnetic field produced by it will be (as seen by observer):
(1) Clockwise
(2) Anti-clockwise
(3) In the direction of motion of charge
(4) In the direction opposite to the motion of charge

19. Four identical rectangular plates with length $l = 2$ m and breadth $b = \frac{3}{2}$m are arranged as shown. If equivalent capacitance for the arrangement is $C_{eq} = \frac{x\epsilon_0}{d}$, the value of x (nearest integer) is: (All units are in SI unit)



(1) 1 (2) 2
(3) 3 (4) 4

20. A 0.06 kg wooden block is dropped from a 150 m tall tower. Simultaneously, a 0.04 kg bullet is fired vertically upwards from the ground at 150 m s^{-1}. The bullet embeds perfectly into the block. Determine the maximum height reached by the combined mass *above* the tower's top. ($g = 10$ ms^{-2})
(1) 80 m (2) 150 m
(3) 120 m (4) 100 m

21. A 20 g bead slides from rest along a circular wire of radius $\frac{10}{\pi}$ cm under constant tangential acceleration. If its kinetic energy reaches 10 mJ exactly upon completing its 5th revolution, find the magnitude of this tangential acceleration.
(1) 0.25 m/s^2 (2) 0.5 m/s^2
(3) 1.0 m/s^2 (4) 0.1 m/s^2

22. A standard bar magnet has a magnetic moment M and pole strength m. If it is sliced longitudinally along its axis into three identical thin magnets, what will be the magnetic moment and pole strength of each individual slice?
(1) $\frac{M}{3}, m$ (2) $\frac{M}{3}, \frac{m}{3}$
(3) $M, \frac{m}{3}$ (4) $\frac{M}{9}, \frac{m}{3}$

23. At what altitude above the Earth's surface does the acceleration due to gravity diminish by exactly 3% of its surface value? (Assume Earth's radius $R = 6400$ km and the height $h \ll R$).
(1) 96 km (2) 192 km
(3) 64 km (4) 128 km

24. A highly elastic wire has a Poisson's ratio of 0.4. Upon applying a longitudinal stretching force, its cross-sectional area decreases uniformly by 2.4%. What is the corresponding percentage elongation of the wire?
(1) 1.5% (2) 3.0%
(3) 4.8% (4) 6.0%

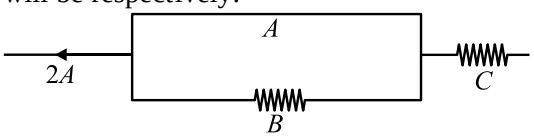
25. An infinitely long straight wire carries a uniform linear charge. It produces an electric field of 3.6×10^5 N/C at a radial distance of 5 cm. Determine the linear charge density of the wire.
(1) 2μC/m (2) 0.5μC/m
(3) 1μC/m (4) 5μC/m

26. A projectile launched vertically upwards crosses a height of 10 m twice, with a time gap of 4 s between the two crossings. Assuming $g = 10$ m/s^2, what is its total time of flight?
(1) $\sqrt{20}$ s (2) $\sqrt{32}$ s
(3) $\sqrt{24}$ s (4) $\sqrt{40}$ s

27. A rectangular glass slab of thickness 12 cm has its rear face silvered. An object is placed 15 cm in front of the unsilvered face. If the final image appears to be 19 cm behind the physical silvered face, calculate the refractive index of the glass.
(1) 1.33 (2) 1.4
(3) 1.5 (4) 1.6

28. What is the electric flux through a cube of side a if a point charge q is placed at the center of one of its edges? (ε_0 is permittivity in free space)
(1) $\frac{q}{\varepsilon_0}$ (2) $\frac{q}{2\varepsilon_0}$
(3) $\frac{q}{4\varepsilon_0}$ (4) $\frac{q}{8\varepsilon_0}$

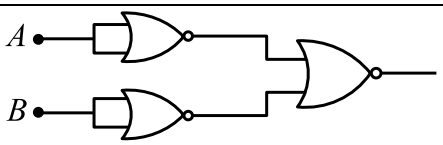
29. A part of circuit is shown in figure. The wire A is very thick copper and wire B and C are identical wires of manganin. The currents in wires A, B, C will be respectively:



(1) 1 A,1 A,2 A (2) 0,2 A,2 A
(3) 2 A,0,2 A (4) 2 A,2 A,2 A

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30.



In the given logic gate combination, for which of the following input values, the output will be 1 (High):

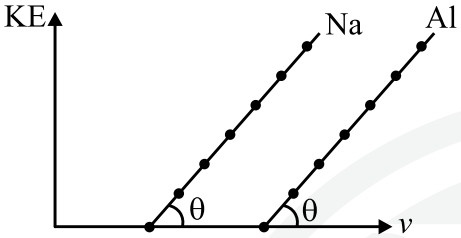
(1) $A = 0, B = 1$

(2) $A = 1, B = 0$

(3) $A = 0, B = 0$

(4) $A = 1, B = 1$

31. From the figure describing photoelectric effect we may infer correctly that:



(1) Na and Al both have the same threshold frequency

(2) Maximum kinetic energy for both the metals depend linearly on the frequency

(3) The stopping potentials are different for Na and Al for the same change in frequency

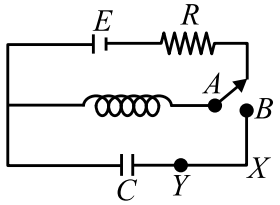
(4) Al is better photosensitive material than Na

32. Two particles A and B of equal masses are suspended from two massless springs of constants K_1 and K_2 respectively. If the maximum velocities during oscillations are equal, the ratio of the amplitude of A and B is:

(1) $\frac{K_1}{K_2}$ (2) $\sqrt{\frac{K_1}{K_2}}$

(3) $\sqrt{\frac{K_2}{K_1}}$ (4) $\frac{K_2}{K_1}$

33. Switch is in position A for long time. At time $t = 0$ it is shifted to position B. Find the maximum charge that will accumulate on capacitor.



(1) $\left(\sqrt{LC}\right)\frac{E}{R}$ (2) $\frac{E}{\left(\sqrt{LC}\right)R}$

(3) $\left(\sqrt{\frac{L}{C}}\right)\frac{E}{R}$ (4) $\left(\sqrt{\frac{C}{L}}\right)\frac{E}{R}$

34. If the angular velocity of the earth's spin is increased such that the bodies at the equator start floating, the duration of the day would be approximately: [Take $g = 9.8\ \text{ms}^{-2}$, radius $R = 6400 \times 10^3$ m, $\pi = 3.14$]

(1) 60 minutes

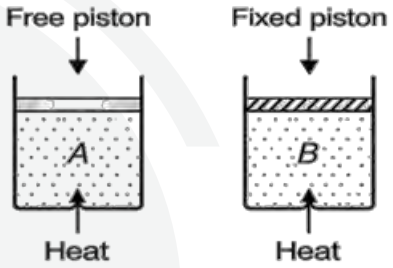
(2) does not change

(3) 84.6 minutes

(4) 1200 minutes

35. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Two cylinders contain the same amount of ideal monatomic gas. If the same amount of heat is given to both, and the temperature rise in cylinder A (free piston) is T_0, then the temperature rise in cylinder B (fixed piston) will be $\frac{5}{3}T_0$.



Reason (R): For an ideal monatomic gas, the molar heat capacity at constant pressure (C_P) is $\frac{5}{3}$ times the molar heat capacity at constant volume (C_V).

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

36. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0\hat{j}$. If an electric field $\vec{E} = E_0\hat{i}$ and magnetic field $\vec{B} = B_0\hat{i}$ act on the particle, its speed will double after a time:

(1) $\frac{2mv_0}{qE_0}$

(2) $\frac{3mv_0}{qE_0}$

(3) $\frac{\sqrt{2}mv_0}{qE_0}$

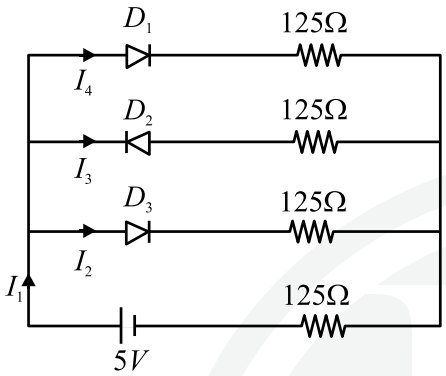
(4) $\frac{\sqrt{3}mv_0}{qE_0}$

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37. Let E represent energy, m represent mass, L represent angular momentum, and G represent the universal gravitational constant. Which of the following physical quantities has the same dimensional formula as the expression $\left[\frac{EL^2}{m^4G^2}\right]$?

(1) Length (2) Time
(3) Mass (4) Angle

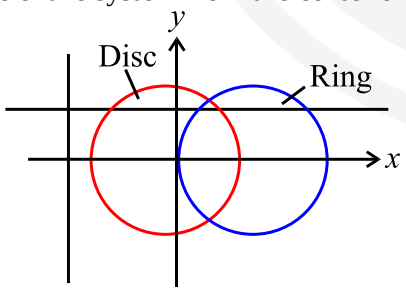
38. If each diode has a forward bias resistance of 25 Ω in the below circuit,



Which of the following options is **correct?**

(1) $\frac{I_3}{I_4}=1$ (2) $\frac{I_2}{I_3}=1$
(3) $\frac{I_1}{I_2}=1$ (4) $\frac{I_1}{I_2}=2$

39. A uniform thin ring of mass M and radius R is placed on a uniform thin disc of mass $2M$ and radius R such that the ring passes through the center of the disc. The distance of the center of mass of the entire system from the center of the disc is:

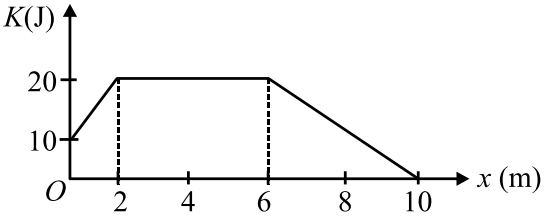


(1) $\frac{R}{2}$ (2) $\frac{R}{3}$
(3) $\frac{2R}{3}$ (4) $\frac{R}{4}$

40. A string firmly fixed at both ends oscillates in its 5th harmonic. The profile of the standing wave is given by $y=0.4\sin(0.314x)\cos(300\pi t)$, where all parameters are in SI units. Calculate the total length of the string.

(1) 40 m (2) 25 m
(3) 50 m (4) 60 m

41. The kinetic energy K of a particle moving along the x-axis varies with its position (x) as shown in the figure. The magnitude of force acting on the particle at $x=9$ m is:



(1) Zero (2) 5 N
(3) 20 N (4) 7.5 N

42. In a vacuum, a plane electromagnetic wave possesses an energy density of $\frac{2}{\pi}\times10^{-7}$ J m^{-3}. Determine the peak amplitude of its magnetic field.

(1) 200 nT
(2) 400 nT
(3) 600 nT
(4) 800 nT

43. A massless box contains an ideal gas (molar mass M, adiabatic index γ) and travels at a constant velocity of $3v$ on a frictionless horizontal surface. If the box is suddenly brought to a complete halt, what will be the resulting rise in the temperature of the gas? (R is universal gas constant)

(1) $\frac{3Mv^2}{(\gamma-1)R}$ (2) $\frac{3(\gamma-1)Mv^2}{2R}$
(3) $\frac{9\gamma Mv^2}{2R}$ (4) $\frac{9(\gamma-1)Mv^2}{2R}$

44. A planet has double the mass of the earth. Its average density is equal to that of the earth. An object weighing W on earth will weigh on that planet:

(1) $2^{\frac{2}{3}}W$ (2) W
(3) $2^{\frac{1}{3}}W$ (4) $2W$

45. A tuning fork of frequency 440 Hz produces 6 beats/sec when sounded together with a vibrating guitar string. When the tension in the guitar string is slightly increased, the beat frequency is observed to increase to 8 beats/sec. What was the original frequency of the guitar string?

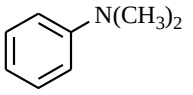
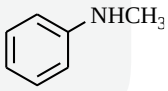
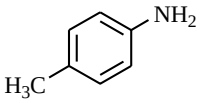
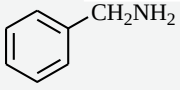
(1) 434 Hz
(2) 446 Hz
(3) 448 Hz
(4) 432 Hz

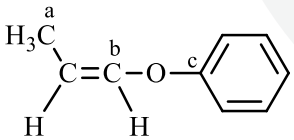
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46. The element having the highest first ionization enthalpy is:
(1) P (2) Al
(3) N (4) C

47. A compound A with molar weight of 80 g is dissolved in a solvent having density of 0.4 g mol^{-1}. Assuming no change in volume upon dissolution, the molality (in mol/kg) of a 3.2 molar solution is:
(1) 4 (2) 8
(3) 2 (4) 16

48. Metallic sodium does **not** react normally with:
(1) Gaseous ammonia
(2) Tert-butyl alcohol
(3) But-2-yne
(4) Ethyne

49. Amongst the compounds given, the one that would form a brilliant coloured dye on treatment with $NaNO_2$ in dil. HCl followed by addition to an alkaline solution of phenol is:
(1) 
(2) 
(3) 
(4) 

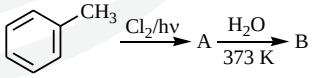
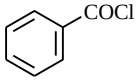
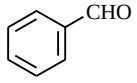
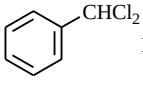
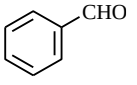
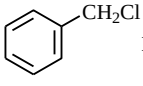
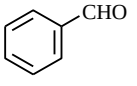
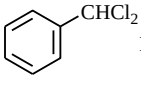
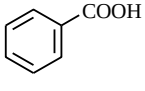
50. In the following molecule,

Hybridisation of carbon a, b and c respectively are:
(1) sp^3, sp^2, sp^2 (2) sp^3, sp, sp
(3) sp^3, sp, sp^2 (4) sp^3, sp^2, sp

51. The **correct** order of basic nature on aqueous solution for the bases
$NH_3, H_2N-NH_2, CH_3CH_2NH_2, (CH_3CH_2)_2NH$ and $(CH_3CH_2)_3N$ is:
(1) $NH_3 < H_2N–NH_2 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$
(2) $NH_3 < H_2N–NH_2 < CH_3CH_2NH_2 < (CH_3CH_2)_2NH < (CH_3CH_2)_2NH$
(3) $H_2N–NH_2 < NH_3 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$
(4) $H_2N–NH_2 < NH_3 < CH_3CH_2NH_2 < (CH_3CH_2)_3N < (CH_3CH_2)_2NH$

52. Identify the **correct** statement(s):
(A) –CN and $–SO_3H$ are meta directing group.
(B) Halides are activating groups.
(C) Activating groups act as ortho- and para directing groups.
(D) $–OCH_3$ and $–NHCOCH_3$ are activating group.
(E) –CN and –OH are meta directing group.
Choose the **correct** answer:
(1) A, C and D only (2) D only
(3) A and D only (4) A, B and E only

53. The compound listed below that contain a –COOH group is:
(1) Picric acid (2) Sulphanilic acid
(3) Carbolic acid (4) Aspirin

54. Identify the **correct** statements about p-block elements and their compounds.
(A) Non-metals generally have lower electronegativity than metals.
(B) Non-metals generally have high ionization enthalpy than metals.
(C) Compounds formed between highly reactive metals and highly reactive non-metals are generally covalent.
(D) Oxides of non-metals are generally basic in nature.
(E) Oxides of metals are generally basic in nature.
(1) A and C only (2) D and E only
(3) B and E only (4) B and D only

55. Identify A and B in the following sequence of reaction.

(1) A =  B = 
(2) A =  B = 
(3) A =  B = 
(4) A =  B = 

56. For the first four members of group-17, which property shows a regular trend down the group?
(A) Electron affinity (B) First ionization energy
(C) Covalent radius (D) Electronegativity
Choose the **correct** answer:
(1) B, C and D only (2) A only
(3) A and B only (4) B only

[Contd...

57. Identify the **correct** statements:
(A) Benzenesulfonyl chloride is known as Hinsberg's reagent.
(B) Secondary and tertiary amines both give carbylamines test.
(C) Tertiary amines react with benzenesulfonyl chloride very easily.
Choose the **correct** answer:
(1) A and C only (2) B and C only
(3) A only (4) A, B and C only

58. Given below are two statements:
Statement I: Pure aniline and other arylamines are generally colourless.
Statement II: Arylamines become coloured on storage because of atmospheric oxidation.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement-I and Statement-II are correct
(2) Both Statement-I and Statement-II are incorrect
(3) Statement-I is correct but Statement-II is incorrect
(4) Statement-I is incorrect but Statement-II is correct

59. The carbohydrate unit associated with DNA is asked about through these statements:
A. It is a hexose sugar
B. It is present in pyranose form
C. It is in "D" configuration
D. It is in α-anomeric form
Choose the **correct** answer:
(1) A, C and D only (2) B, D only
(3) A, B only (4) C only

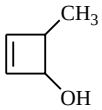
60. Among SbF_5, PCl_5, BrF_5 and $XeOF_4$, which species show octahedral electron-pair arrangement around the central atom?
(1) SbF_5 and PCl_5 (2) BrF_5 and $XeOF_4$
(3) PCl_5 and $XeOF_4$ (4) SbF_5 and BrF_5

61. Hydrolysis of proteins gives amino acids in which the $-NH_2$ group is attached to the carbon atom:
(1) β to –COOH (2) α to –COOH
(3) γ to –COOH (4) δ to –COOH

62. Among Sc, Pb, Na and Te, which element is neither a typical metal nor a nonmetal?
(1) Na (2) Sc
(3) Te (4) Pb

63. Consider the ground state of Mn atom (Z = 25). The numbers of electrons with azimuthal quantum numbers $\ell = 1$ and 2 are, respectively:
(1) 12 and 5 (2) 12 and 4
(3) 16 and 5 (4) 16 and 4

64. Which among the following is an extensive property of a system?
(1) Pressure (2) Volume
(3) Density (4) Refractive index

65. The IUPAC name of  is:
(1) 4-Methylcyclobut-1-en-3-ol
(2) 2-Methylcyclobut-3-en-1-ol
(3) 2-Methylcyclobut-1-en-3-ol
(4) 4-Methylcyclobut-2-en-1-ol

66. Which pair from the following will **not** form an ideal solution?
(1) n-Hexane + n-Heptane
(2) Benzene + Toluene
(3) Water + Ethanol
(4) Bromoethane + Chloroethane

67. The **correct** order of atomic radii in group 13 elements is:
(1) B < Al < Ga < In < Tl
(2) B < Ga < Al < In < Tl
(3) B < Al < In < Ga < Tl
(4) B < Ga < Al < Tl < In

68. Which of the following pairs of a chemical reaction is certain to result in a spontaneous reaction?
(1) Endothermic and decreasing disorder
(2) Exothermic and increasing disorder
(3) Endothermic and increasing disorder
(4) Exothermic and decreasing disorder

69. The half-life of a first order reaction is 15 min. If initial concentration is 0.16 mol L^{-1} and concentration later becomes 0.02 mol L^{-1}, then time is:
(1) 15 min (2) 30 min
(3) 45 min (4) 60 min

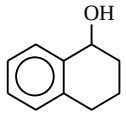
70. Total number of stereoisomers of the following compound is:
$H_3C-\underset{OH}{CH}-\underset{Br}{CH}-CH_3$
(1) 2 (2) 4
(3) 6 (4) 8

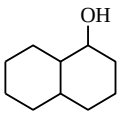
71. Which one of the following oxides is acidic in nature?
(1) PbO (2) SnO_2
(3) CO_2 (4) Na_2O

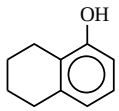
72. Which of the following sulphides is yellow in colour?
(1) ZnS (2) CdS
(3) FeS (4) PbS

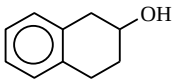
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73. Which alcohol is most reactive towards dehydration of alcohol in acid catalyzed reaction?

(1) 

(2) 

(3) 

(4) 

74. Which of the following statements are **incorrect**?

(A) The electronic configuration of Cu is $[Ar]3d^9 4s^2$.

(B) The magnetic quantum number may have a negative value.

(C) In the ground state, electrons fill orbitals in order of increasing energy.

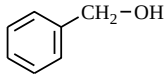
(D) The total number of nodes is equal to $n + \ell$.

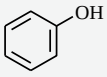
Choose the *most appropriate* answer from the options given below:

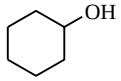
(1) A, C and D only (2) A and D only

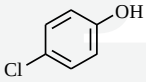
(3) A, B and C only (4) B, C and D only

75. Which of the following is most acidic?

(1) 

(2) 

(3) 

(4) 

76. The electronic configuration of Au (atomic number 79) is:

(1) $[Xe]4f^{14}5d^9 6s^2$ (2) $[Xe]4f^{14}5d^{10}6s^1$

(3) $[Xe]4f^{14}5d^8 6s^2$ (4) $[Kr]4d^{10}5s^1$

77. Match **List-I** with **List-II**.

List-I (Cell)		**List-II (Use/Property/Reaction)**	
(A)	Leclanche cell	(I)	Converts energy of combustion into electrical energy
(B)	Ni-Cd cell	(II)	Does not involve any ion in solution and is used in hearing aids
(C)	Fuel cell	(III)	Rechargeable
(D)	Mercury cell	(IV)	Reaction at anode $Zn \rightarrow Zn^{2+} + 2e^-$

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV

(2) A-III, B-I, C-IV, D-II

(3) A-IV, B-III, C-I, D-II

(4) A-II, B-III, C-IV, D-I

78. Henry's law constant for O_2 in water at 298 K is 2.5×10^4 bar. At pressure 0.05 bar, mole fraction solubility is:

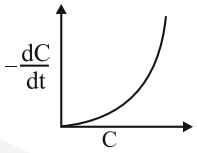
(1) 2×10^{-6}

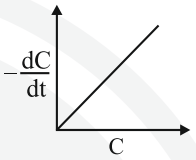
(2) 5×10^{-6}

(3) 2×10^{-5}

(4) 5×10^{-5}

79. In two different reactions, involving a single reactant in each case, a plot of rate of reaction on the y-axis, versus concentration of the reactant on the x-axis, yields two different curves shown below.

(I) 

(II) 

What is the possible orders of the reactions (I), (II)?

(1) 1, 2

(2) 2, 1

(3) 1, 1

(4) 0, 1

80. What pressure of H_2 would be required to make the emf of hydrogen electrode zero in pure water at 25°C?

(1) 10^{-14} bar

(2) 10^{-7} bar

(3) 1 bar

(4) 10^{-1} bar

81. What is value of (P + Q), where P is the n-factor of $Ca(OH)_2$ and Q is the n-factor of HCl?

(1) 5 (2) 4

(3) 3 (4) 2

82. Given below are two statements:

Statement I: Enzymes are not specific in their action.

Statement II: Enzymes are required in small amount.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.

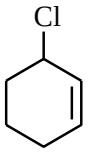
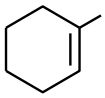
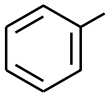
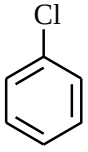
(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

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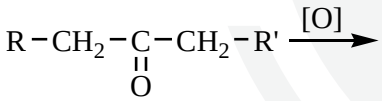
83. Match **List I** with **List II**:

List I (Structure)		List II (Type of halide)	
A.		I.	Vinylic halide
B.		II.	Benzylic halide
C.		III.	Aryl halide
D.		IV.	Allylic halide

Choose the **correct** answer from the options given below:

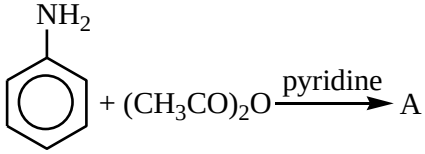
(1) A-IV, B-I, C-II, D-III
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-II, C-IV, D-III
(4) A-II, B-III, C-I, D-IV

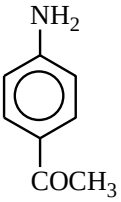
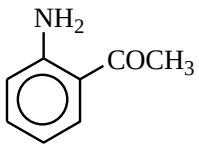
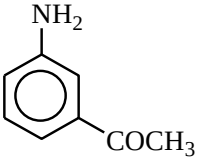
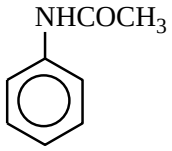
84. What are the possible products formed by oxidation of the given compound with strong oxidising agent and at elevated temperatures?



(1) R–COOH and R′–CH_2COOH
(2) R–CHO and R′–CH_2COOH
(3) R–COOH and R′–CH_2CHO
(4) R–CHO and R′–CH_2CHO

85. The major product A of the given reaction is:



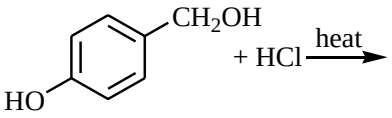
(1) 
(2) 
(3) 
(4) 

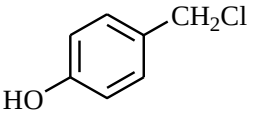
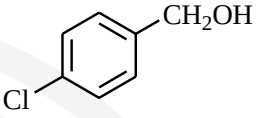
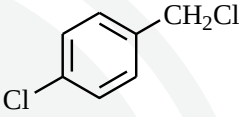
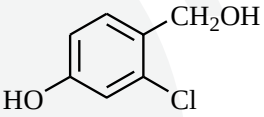
86. The given reaction is an example of:

$CuSO_4(aq) + Zn(s) \rightarrow Cu(s) + ZnSO_4(aq)$

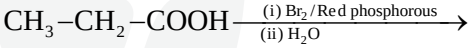
(1) Metal displacement reaction
(2) Non-metal displacement reaction
(3) Decomposition reaction
(4) Combination reaction

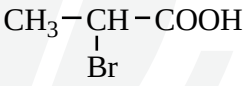
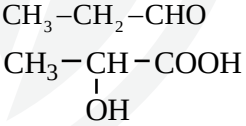
87. The product obtained in the given reaction is:

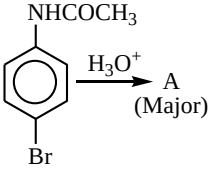


(1) 
(2) 
(3) 
(4) 

88. The product obtained in the following reaction is;



(1) 
(2) CH_3-CH_2-CHO
(3) 
(4) $CH_2=CH-COOH$

89. 

The product A is:

(1) 4-Bromoaniline (2) 4-Bromophenol
(3) Bromobenzene (4) 4-Aminophenol

90. For a reaction,

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$;

Identify dihydrogen (H_2) as a limiting reagent in the following reaction mixtures.

(1) 14 g of N_2 + 4 g of H_2
(2) 28 g of N_2 + 7 g of H_2
(3) 56 g of N_2 + 10 g of H_2
(4) 35 g of N_2 + 8 g of H_2

91. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): A test cross of F_1 progeny of snapdragon (being studied for inheritance of flower colour) gives a 1:1 ratio of red flowered and white flowered plants.

Reason (R): A test cross can easily be analysed to predict the genotype of the test organism.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

92. Choose the **incorrect** statement for bacteria that produce large amounts of methane along with CO_2 and H_2 from cellulosic material.

(1) They are found in gut of ruminants.
(2) They are found naturally in sewage water.
(3) They grow aerobically on cellulosic material.
(4) They help in production of gobar gas.

93. Match **List-I** with **List-II**.

List-I (Organism)		**List-II (Characteristic)**	
(A)	*Chara*	(I)	Oogamous sexual reproduction, chloroplasts contain chlorophyll *a* and *d*
(B)	*Anabaena*	(II)	Stored food is laminarin or mannitol, chloroplasts contain chlorophyll *a* and *c*
(C)	*Polysiphonia*	(III)	Chloroplasts contain pyrenoids, chlorophyll *a* and *b*
(D)	*Dictyota*	(IV)	Chloroplasts are not present, can fix atmospheric nitrogen in specialised cells

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-IV, D-III

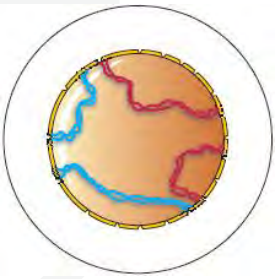
94. Identify the **incorrect** statement.

(1) The species continue to live through millions of years unless threatened by natural or anthropogenic extinction.
(2) Sexual reproduction enhances the survival advantage of species.
(3) Reproduction becomes a vital process without which species cannot survive for long.
(4) Sexual reproduction leads to formation of clones of parents.

95. Identify the similarity between *Euglena* and Venus fly trap.

(1) Possess photosynthetic pigments and cell wall.
(2) Can photosynthesise as well as can behave as heterotroph.
(3) Are eukaryotic and multicellular.
(4) Show locomotion and are aquatic.

96. Go through the stage of mitosis in the cell shown below and choose the **correct** statement about it.



(1) It is one of the sub stages of the resting phase of cell cycle.
(2) It is the last stage of the most dramatic phase of cell cycle.
(3) The DNA has already replicated but the daughter chromosomes are yet to be formed.
(4) This cell will undergo cytokinesis by the formation of cell plate.

97. The number of amphibian species;

(1) is higher in Eastern Ghats as compared to Western ghats.
(2) is decreasing because they appear to be more vulnerable to extinction than other animal groups.
(3) is more than the combined total of the species of fishes, fungi, reptiles and mammals.
(4) in Amazon rain forests is approximately 4270.

[Contd...

98. Consider the following statements regarding the cell cycle:
A. The interphase lasts more than 95% of the duration of the cell cycle.
B. In the S phase, the chromosome number in a cell doubles.
C. The G_2 phase corresponds to the interval between mitosis and initiation of DNA replication.
D. In the quiescent stage (G_0), cells are metabolically inactive.
E. The M Phase starts with cytokinesis and ends with karyokinesis.
Choose the option with the **correct** statement(s).
(1) A only (2) B and E
(3) E only (4) A, B and C

99. Identify the **incorrectly** matched pair.
(1) Pteridophytes - Main plant body is a sporophyte
(2) Gymnosperms - Ovules are not enclosed by any ovary wall
(3) Bryophytes - Possess well-differentiated vascular tissues
(4) Angiosperms - Seeds are enclosed in fruits

100. The **correct** sequence of components of a forest ecosystem from top to bottom strata is;
(1) Herbs and Grasses → Shrubs → Trees
(2) Herbs and Grasses → Trees → Shrubs
(3) Shrubs → Herbs and Grasses → Trees
(4) Trees → Shrubs → Herbs and Grasses

101. Two leaves A and B have surface area 50 cm^2 and 100 cm^2, respectively. They both increase their surface area by 5 cm^2 in a day. The absolute growth rate;
(1) of A is more than B.
(2) of B is more than A.
(3) is 5 cm^2 per day.
(4) is 0.1 and 0.05 for leaf A and B, respectively.

102. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Conjoint open vascular bundles	(I)	Dicot root
(B)	Radial open vascular bundles	(II)	Monocot stem
(C)	Conjoint closed vascular bundles	(III)	Monocot root
(D)	Radial closed vascular bundles	(IV)	Dicot stem

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

103. Given below are two statements:
Statement I: The pattern of inheritance of phenylketonuria cannot be studied by pedigree analysis.
Statement II: ABO blood group system in humans is a case of polygenic inheritance.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

104. Which of the following is a function of the mesosome in a prokaryotic cell?
(1) Synthesis of proteins
(2) Trapping light energy
(3) DNA replication and distribution to daughter cells
(4) Motility

105. The curly top virus spreads through a plant via the tissue that is;
(1) made of similar type of cells.
(2) made of only dead cells.
(3) responsible for carbon assimilation.
(4) a part of conducting tissue system.

106. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Schleiden & Schwann	(I)	*Omnis cellula-e cellula*
(B)	Rudolf Virchow	(II)	Fluid mosaic model of plasma membrane
(C)	Singer & Nicolson	(III)	Cell theory formulation
(D)	Robert Brown	(IV)	Discovered the nucleus

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

107. During the light reaction, FNR is;
(1) active during non-cyclic photophosphorylation only.
(2) the enzyme that catalyses reduction of $NADPH + H^+$.
(3) present in the stromal lamellae.
(4) the primary electron acceptor from PS II.

108. Which one of the following has the largest population in a food chain in most ecosystems?
(1) Producers
(2) Primary consumers
(3) Secondary consumers
(4) Primary carnivores

109. Given below are two statements:
Statement I: During transcription, adenosine complements forms base pair with thymine.
Statement II: In transcription, only a segment of DNA and only one of the strands is copied into RNA.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

110. The first stable product of CO_2 fixation in C_4 plants is;
(1) 3-phosphoglyceric acid.
(2) Ribulose bisphosphate.
(3) Oxaloacetic acid.
(4) Malic acid.

111. Select the **correctly** matched pair with respect to fruit.
(1) Pea - Drupe
(2) Maize - Caryopsis
(3) Sunflower - Berry
(4) Coconut - Legume

112. Arrange the following events in correct chronological sequence of occurrence in a typical flowering plant.
A. Formation of the female gametophyte
B. Fruit formation
C. Degradation of antipodals
D. Transfer of pollen to stigma
E. Triple fusion
Choose the **correct** option.
(1) B - A - C - D - E
(2) D - C - B - E - A
(3) C - D - E - A - B
(4) A - D - E - C - B

113. Which of the following is a **correct** statement regarding the rules of Binomial Nomenclature?
(1) The first word denoting the genus starts with a small letter.
(2) The specific epithet starts with a capital letter.
(3) Biological names when handwritten are jointly underlined.
(4) Biological names are generally in Latin and written in italics.

114. The diagram below represents a crucial step in decomposition of detritus being performed by an organism. Choose the **correct** option related to it.



(1) The step shows the breakdown of detritus into smaller particles by detritivores, process known as catabolism.
(2) This step is coupled with release of inorganic nutrients above the soil.
(3) This is the first step of decomposition that occurs on or within the top layers of soil.
(4) The raw material for this process is living plants and animals.

115. Read the following statements regarding the *lac* operon.
A. Lactose is the substrate for the enzyme beta-galactosidase and it regulates switching on and off of the operon.
B. The repressor of the operon is synthesised constitutively from the *z* gene.
C. In the presence of the inducer, the repressor is activated by interaction with it.
D. Regulation of *lac* operon by repressor is referred to as negative regulation.
E. The *y* gene codes for transacetylase.
In the light of above statements choose the **correct** answer from the options given below:
(1) B and E are correct, A, C and D are incorrect
(2) C and D are correct, A, B and E are incorrect
(3) A and D are correct, B, C and E are incorrect
(4) A, C and E are correct, B and D are incorrect

116. How many of the following statements about the anatomy of a dicot stem are **correct**?
A. The vascular bundles are arranged in a ring.
B. The primary xylem has exarch protoxylem.
C. The outermost layer is the epidermis, which may bear trichomes.
D. The cortex is differentiated into hypodermis, cortical layers, and endodermis.
E. The central portion of the stem is occupied by a large, conspicuous pith.
Choose the **correct** option.
(1) Two only
(2) Three only
(3) Four only
(4) Five

117. Match **List-I** with **List-II**.

List-I (Organisms)		List-II (Description of interaction between them)	
(A)	Cuckoo laying eggs in Crow's nest	(I)	One benefitted, another unaffected
(B)	Orchid growing as an epiphyte on a mango branch	(II)	One benefitted, another harmed
(C)	Fig and pollinator wasp	(III)	Both lose
(D)	Visiting Flamingoes and resident fishes in some shallow South American lakes	(IV)	Both benefitted

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-II, D-I

118. Select the **correct** statement regarding fungi.
(1) The cell walls of fungi are composed of peptidoglycan.
(2) With the exception of yeasts, which are multicellular, all other fungi are filamentous.
(3) The network of hyphae is known as mycelium.
(4) Most fungi are autotrophic and absorb soluble organic matter from living substrates.

119. Identify the **correct** statements from the following regarding Human Genome Project (HGP).
A. The total number of genes is estimated at 30,000.
B. More than 20% of the genome codes for proteins.
C. Chromosome 1 has the most genes (2968) and the X chromosome has the fewest (231).
D. The human genome contains 3164.7 million base pairs.
E. Snips are present at about 1.4 million locations.
Choose the **correct** option.
(1) A, D and E (2) A, B and D
(3) B, C and E (4) A, C and D

120. The phytohormone that is a carotenoid derivative is;
(1) auxin. (2) gibberellin.
(3) cytokinin. (4) abscisic acid.

121. Identify the **correct** statement.
(1) In gymnosperms, the male and female gametophytes have an independent existence.
(2) Liverworts reproduce asexually by fragmentation or specialized structures called gemmae.
(3) In mosses, strobili or cones are formed.
(4) All pteridophytes show an event precursor to seed habit.

122. The fruit of strawberry;
(1) has no role of ovary in its formation.
(2) is a true fruit.
(3) has thalamus as the edible part.
(4) develops by apomixis.

123. How many of the following features are seen in mustard plant?

Fibrous root system, Alternate phyllotaxy, Parallel venation, Actinomorphic flower, Superior ovary, Apocarpous condition, Parietal placentation

(1) Four only
(2) Three only
(3) Two only
(4) One only

124. According to the Watson-Crick model of DNA, the two polynucleotide chains have;
(1) 5'→3' polarity in same direction.
(2) phosphate groups at the same end.
(3) antiparallel polarity.
(4) exactly same sequence of nitrogenous bases.

125. In a dorsiventral leaf, the vascular bundles are surrounded by a layer of thick-walled cells. These cells are known as:
(1) Mesophyll cells.
(2) Bundle sheath cells.
(3) Subsidiary cells.
(4) Companion cells.

126. Identify the **incorrect** statement about Down's syndrome.
(1) It is a case of trisomy.
(2) The affected individual suffers from congenital heart disease.
(3) The individual exhibits overall masculine development along with gynaecomastia.
(4) Physical, psychomotor, and mental development is retarded.

127. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Glucose	(I)	Have RQ below 1
(B)	Fats	(II)	Will degrade into amino acids for respiration
(C)	Proteins	(III)	Favored substrate for respiration
(D)	Pyruvic acid	(IV)	Can enter fermentation as well as Krebs' cycle depending upon cellular need

Choose the *most appropriate* answer from the options given below.

(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-III, B-I, C-IV, D-II
(4) A-I, B-III, C-II, D-IV

128. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Plum	(I)	Polyadelphous stamens
(B)	*Citrus*	(II)	Half-inferior ovary
(C)	Canna	(III)	Imbricate aestivation
(D)	*Cassia*	(IV)	Asymmetrical flower

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-I, C-II, D-IV

129. The "bakanae" (foolish seedling) disease of rice seedlings is caused by a:

(1) Bacterium. (2) Virus.
(3) Fungus. (4) Viroid.

130. Given below are two statements:

Statement I: In an expanding population, the number of pre-reproductive individuals is more than that of reproductive and post-reproductive individuals.

Statement II: Sex ratio is a population attribute.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

131. How many of the following organisms are not placed under Kingdom Protista?

A. Desmids
B. *Amoeba*
C. *Paramoecium*
D. *Spirogyra*
E. Slime moulds

Choose the **correct** option.

(1) Three
(2) Four
(3) Five
(4) One

132. The 'endomembrane system' of a eukaryotic cell includes;

(1) Endoplasmic reticulum, Golgi complex, Lysosomes, and Vacuoles.
(2) Endoplasmic reticulum, Mitochondria, Ribosomes, and Lysosomes.
(3) Golgi complex, Mitochondria, Chloroplasts, and Vacuoles.
(4) Endoplasmic reticulum, Peroxisomes, Lysosomes, and Golgi complex.

133. The DNA can be digested by the enzyme:

(1) DNA polymerase.
(2) DNA ligase.
(3) RNase.
(4) DNase.

134. Select the **correct** statement from the following.

(1) In isobilateral leaves, the stomata are present only on the abaxial epidermis.
(2) All tissues except epidermis and vascular bundles constitute the stele.
(3) The endodermis of dicot stem is also referred to as the starch sheath.
(4) In monocot roots, the pith is small and inconspicuous.

135. Considering both substrate level phosphorylation, and oxidative phosphorylation of NADH + H^+ and $FADH_2$ produced from one turn of Krebs' cycle, a total of how many ATP will be synthesised?

(1) 2 only
(2) 12 only
(3) up to 38
(4) 0

136. Variation in gene frequencies within populations can occur by chance rather than by natural selection. This is referred to as:
(1) Genetic drift (2) Random mating
(3) Genetic load (4) Genetic flow

137. Which of the following is **correct** regarding the specialized connective tissue "Bone"?
(1) The bone cells (osteocytes) are present in spaces called lamellae.
(2) Bone lacks a hard and non-pliable ground substance.
(3) It is the main tissue that provides structural frame to the body.
(4) Bone cells are rich in chondroitin salts which provide strength.

138. Match **List-I** with **List-II**.

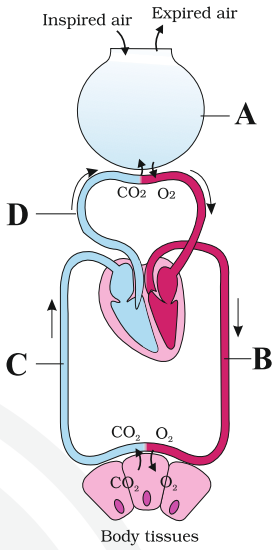
List-I (Secondary metabolite)		**List-II (Category)**	
(A)	Vinblastine	(I)	Alkaloid
(B)	Anthocyanin	(II)	Toxin
(C)	Abrin	(III)	Drug
(D)	Morphine	(IV)	Pigment

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-I, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-IV, C-II, D-I

139. Choose the **correct** statement regarding the human respiratory system:
(1) The conducting part extends from external nostrils up to the terminal bronchioles.
(2) Expiration occurs when intra-pulmonary pressure is lower than atmospheric pressure.
(3) The trachea, primary, secondary, and tertiary bronchi are supported by complete cartilaginous rings.
(4) Vital capacity is the sum of IRV + ERV + RV.

140. Identify the **correct** statement regarding Malignant Tumors:
(1) They show the property of contact inhibition, reaching distant sites via blood
(2) They show the property of metastasis, reaching distant sites via blood.
(3) They grow very slowly and cause little damage.
(4) They are non-cancerous in nature.

141. In the diagram showing the exchange of gases at the alveolus and the body tissues with blood and transport of carbon dioxide and oxygen, what does the pO_2 and pCO_2 (in mm Hg) represent in blood vessel marked as B?



(1) $pO_2 = 40$, $pCO_2 = 45$
(2) $pO_2 = 95$, $pCO_2 = 40$
(3) $pO_2 = 104$, $pCO_2 = 40$
(4) $pO_2 = 40$, $pCO_2 = 40$

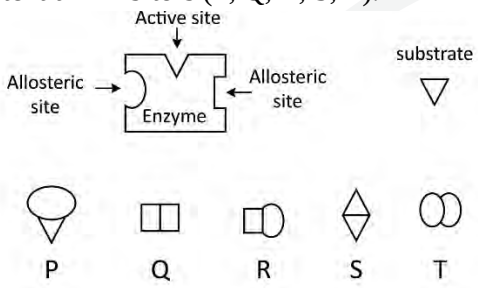
142. Which of the following is responsible for the lub sound of the heart?
(1) Closure of the semilunar valves at the beginning of ventricular diastole.
(2) Closure of the tricuspid and bicuspid valves at the end of ventricular systole.
(3) Closure of the tricuspid and bicuspid valves at the beginning of ventricular systole..
(4) Blood hitting the wall of the ventricles during rapid filling.

143. The glomerular filtration rate is **not** affected by:
(1) increased dietary uptake of water and fluids.
(2) total volume of the blood flowing per minute through the Bowman's capsule.
(3) increased Renin production from Juxta glomerular cells.
(4) increased supply of nutrients to the blood in afferent arteriole.

144. Choose the **correctly** paired match from the following:
(1) *Petromyzon* — Hagfish
(2) *Pristis* — Saw fish
(3) *Carcharodo*n — Dog fish
(4) *Exocoetus* — Fighting fish

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The development of *Periplaneta americana* is paurometabolous.
Reason R: The nymphal stage grows by moulting about 13 times to reach the adult form.
In the light of the above statements, choose the **correct** answer from the options given below.
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

146. The diagram represents an enzyme, its substrate and potential inhibitors (P, Q, R, S, T).



Which one of the following combinations is the **best** pair of competitive inhibitors for the enzyme?
(1) P, S (2) Q, R
(3) S, T (4) R, T

147. Choose the **incorrect** statement:
(1) Persons with 'O' blood group have both anti-A and anti-B antibodies in plasma.
(2) *Erythroblastosis foetalis* can occur if the mother is Rh positive and the fetus is Rh negative.
(3) Serum is plasma without clotting factors.
(4) Fibrinogens are needed for clotting or coagulation of blood.

148. Match **List-I** with **List-II.**

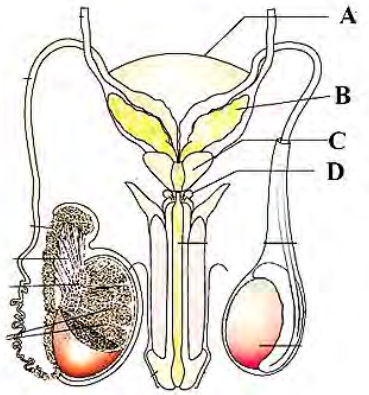
List-I (Animal)		**List-II ((Specialised feature)**	
(A)	*Physalia*	(I)	Calcareous ossicles
(B)	*Spongilla*	(II)	Chitinous exoskeleton
(C)	*Apis*	(III)	Spicules
(D)	*Asterias*	(IV)	Cnidoblast

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-II, D-I

149. Select the **correct** statement regarding 'Saheli', a contraceptive pill.
A. It is an IUD used by females.
B. It is a 'once-a-week' pill with very few side effects.
C. It contains a high concentration of estrogen.
D. It was developed by the Indian Council of Medical Research (ICMR).
(1) A and B
(2) B and D
(3) D only
(4) B only

150. Consider the following statements regarding proteins and select the **correct** combination:
A. The first amino acid in a polypeptide chain is called the C-terminal amino acid.
B. Adult human hemoglobin consists of 4 subunits—two alpha types and two beta types.
C. In a proteins' tertiary structure, the peptide bonds are the only bonds stabilizing the fold.
D. Collagen is the most abundant protein in the entire biosphere.
E. The secondary structure of a protein is strictly determined by disulfide linkages.
(1) B and D only (2) A, B, and D
(3) B, C, and E (4) B only

151. Observe the diagram of the male reproductive system. Which of the following labelled parts represents seminal vesicles?



(1) A (2) B
(3) C (4) D

152. Which of the following is a copper-releasing IUD?
(1) Progestasert (2) Multiload 375
(3) Lippes loop (4) LNG-20

153. Marijuana is a drug obtained from the:
(1) Latex of *Papaver somniferum*
(2) Leaves and inflorescence of *Cannabis sativa*
(3) Flowers of *Datura*
(4) Fruits of *Erythroxylum coca*

[Contd...

154. Choose the **incorrect** statement regarding Downstream processing:
(1) It includes separation and purification of the product.
(2) The product is formulated with suitable preservatives.
(3) It occurs before the biosynthetic stage in a bioreactor.
(4) Quality control testing for each product is required.

155. Regarding the characteristics of Phylum Aschelminthes, consider the following statements:
A. They are often called roundworms because their body is circular in cross-section.
B. An excretory tube removes body wastes from the body cavity through the excretory pore.
C. The alimentary canal is complete with a well-developed muscular pharynx.
D. They are dioecious, and often males are longer than females.
E. Fertilization is typically external, and development is always direct.
Which of the above statements are **correct**?
(1) A, B, C and E (2) A, C, D and E
(3) A, B and C only (4) B, C and D only

156. Which of the following cells act as an HIV factory in the human body?
(1) B-lymphocytes (2) Helper T-cells
(3) Macrophages (4) Cytotoxic T-cells

157. Choose the **correct** statement regarding the human excretory system:
(1) The human kidney can produce urine nearly 10 times concentrated than the initial filtrate.
(2) Presence of glucose and ketone bodies in urine are indicative of Diabetes Mellitus.
(3) Dialysing unit contains a fluid which has the same composition as plasma including nitrogenous wastes.
(4) Sweat produced by sweat glands is primarily meant for excretion of urea.

158. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Trophoblast	(I)	Contains 8 to 16 blastomeres
(B)	Inner cell mass	(II)	Outer layer that gets attached to the endometrium
(C)	Blastocyst	(III)	Differentiates into the embryo
(D)	Morula	(IV)	Arranged into trophoblast and inner cell mass

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-IV, D-II
(2) A-II, B-III, C-I, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-III, B-IV, C-II, D-I

159. Choose the **incorrect** statement:
(1) Transgenic animals can be used to study the contribution of genes in development.
(2) 95% of all existing transgenic animals are mice.
(3) The first transgenic cow, Rosie, produced milk enriched with human alpha-lactalbumin.
(4) α-1-antitrypsin" is used to treat Rheumatoid arthritis.

160. Match **List-I** with **List-II** w.r.t. frog.

List-I		List-II	
(A)	Cloaca	(I)	Present in the kidney of male frogs to help in sperm passage
(B)	Bidder's Canal	(II)	A common chamber used to pass faeces, urine, and sperms to the exterior
(C)	Mesorchium	(III)	A double fold of peritoneum that attaches testes to the kidneys
(D)	Conus Arteriosus	(IV)	A sac-like structure on the ventral side of the heart where the ventricle opens

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

161. A 30-year-old woman has a prolonged luteal phase with two months of missed periods, but is not pregnant. Which of the following hormonal abnormality likely causes the corpus luteum to persist?
(1) Deficient secretion of Progesterone
(2) Continuous hypersecretion of LH
(3) An abrupt preovulatory surge of FSH.
(4) A complete absence of circulating estrogen.

162. Identify the **correctly** matched pair of enzyme class and its reaction type:
(1) Dehydrogenases — Transfer of a group G (other than hydrogen) between a pair of substrates
(2) Hydrolases — Catalyze the linking together of two compounds
(3) Lyases — Removal of groups from substrates by mechanisms other than hydrolysis
(4) Isomerases — Catalyze the transfer of electrons

163. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Adaptive Radiation	(I)	Selection of drug resistant bacteria
(B)	Convergent Evolution	(II)	Wings of butterfly and bird
(C)	Divergent Evolution	(III)	Darwin's Finches
(D)	Anthropogenic Evolution	(IV)	Thorns of *Bougainvillea* and tendrils of *Cucurbita*

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-II, C-IV, D-I

164. Which of the following statements is **correct** regarding the morphology of a cockroach?
(1) The head is formed by the fusion of eight segments and shows great mobility.
(2) In both sexes, the 10th segment bears a pair of jointed filamentous structures called anal styles.
(3) The first pair of wings (tegmina) are transparent and used in flight.
(4) In females, the 7th sternum is boat-shaped and together with the 8th and 9th sterna forms a brood or genital pouch.

165. Complete the Analogy:
Cellulose : Cellulase :: Chitin : _____
(1) Lysozyme (2) Chitinase
(3) Protease (4) Ribonuclease

166. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Innate Immunity	(I)	Memory-based, pathogen-specific
(B)	Acquired Immunity	(II)	Non-specific, present at birth
(C)	IgA antibodies	(III)	Transferred from mother to child during lactation
(D)	Passive Immunity	(IV)	Transfer of preformed antibodies

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-IV, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-III, C-I, D-II

167. A gene is inserted into the *Pvu*II site of the cloning vector (pBR322) and transformed into *E. coli* cells. Which one of the following statements is then **true**?
(1) Recombinants can be selected by plating on ampicillin containing medium.
(2) Recombinants can be selected by plating on tetracycline containing medium.
(3) Recombinants cannot be selected by plating either on ampicillin or tetracycline containing medium.
(4) Recombinants can be selected by plating simultaneously on ampicillin and tetracycline containing medium.

168. A patient has a defective ADA gene. Which of the following is a permanent cure for this condition?
(1) Periodic infusion of genetically engineered lymphocytes.
(2) Enzyme replacement therapy.
(3) Bone marrow transplantation at adult stage.
(4) Introduction of the ADA gene into cells at early embryonic stages.

169. Choose the **incorrect** statement regarding the human skeletal system:
(1) The 8th, 9th, and 10th pairs of ribs are called vertebrochondral ribs.
(2) The floating ribs are not connected ventrally to the sternum.
(3) Each half of the pectoral girdle consists of a clavicle and a scapula.
(4) The human skull is monocondylic.

170. Choose the **incorrect** statement regarding hormones:
(1) Insulin is a peptide hormone that lowers blood glucose levels.
(2) Glucocorticoids stimulate gluconeogenesis, lipolysis, and proteolysis.
(3) Thyrocalcitonin (TCT) increases blood calcium levels.
(4) Epinephrine and Norepinephrine are called emergency hormones or Catecholamines.

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171. Given below are two statements:
Statement I: Cenozoic era is also called the "Age of Mammals,
Statement II: Analogous organs have different origins but similar functions because different organisms evolved in similar habitats/ environments.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

172. Assisted Reproductive Technology, GIFT, involves the transfer of:
(1) Zygote into the fallopian tube.
(2) Embryo into the uterus.
(3) Ovum into the fallopian tube.
(4) Zygote into the uterus.

173. Choose the **correct** statement regarding *Bt* toxin:
(1) The toxin exists as an active toxin in the *Bacillus*.
(2) The inactive protoxin gets converted into active form in the acidic pH of the insect gut.
(3) The activated toxin binds to the surface of midgut epithelial cells.
(4) Bt toxin is a broad-spectrum herbicide.

174. In a neuron, the granular bodies found in the cell body and dendrites are called:
(1) Schwann cells (2) Nodes of Ranvier
(3) Nissl's granules (4) Synaptic vesicles

175. Choose the **correctly** matched pair:
(1) Thermostable DNA polymerase — *Agrobacterium tumefaciens*
(2) *Hind* II — First restriction endonuclease
(3) Undifferentiated mass of cells — Bioreactor
(4) Recognition site — *ori*

176. Which hormone acts on the exocrine pancreas and stimulates the secretion of water and bicarbonate ions?
(1) Gastrin
(2) Secretin
(3) Cholecystokinin (CCK)
(4) GIP

177. Match **List-I** with **List-II**.

List-I (Joint type)		**List-II (Location)**	
(A)	Ball and Socket	(I)	Between Atlas and Axis
(B)	Hinge Joint	(II)	Between Carpal and Metacarpal of thumb
(C)	Pivot Joint	(III)	Between Humerus and Pectoral Girdle
(D)	Saddle Joint	(IV)	Knee Joint

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-I, D-II
(2) A-III, B-IV, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-II, C-I, D-III

178. To protect a crop from a specific herbicide, a scientist wants to use a "Biological Vector" that can naturally deliver genes into plant cells. Which should they choose?
(1) *Escherichia coli*
(2) *Agrobacterium tumefaciens*
(3) *Salmonella typhi*
(4) *Bacillus thuringiensis*

179. Which of the following structures is the first to appear as a result of the differentiation of the inner cell mass immediately after implantation?
(1) Ectoderm and endoderm
(2) Mesoderm and blastocyst
(3) Endoderm and placenta
(4) Trophoblast

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In *Ctenoplana*, digestion is both extracellular and intracellular.
Reason R: In Ctenophores, fertilisation is external with direct development.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

■■■

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